LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Group Travel Expense Simplification Engine

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Architecture Selection: We chose Layered Architecture for the Group Travel Expense Simplification Engine.

Architectural Choice and Reasons: The problem statement describes a multi-currency expense-sharing utility that records uneven group expenses, converts foreign currencies using daily exchange rates, and computes a minimum-transaction debt settlement graph. Layered Architecture separates user interaction from these business operations and from persistent storage. It also keeps related operations such as expense recording, currency normalization and settlement calculation organized behind clear component boundaries without introducing the operational complexity of many independently deployed services.

Security Advantage: The layered structure allows sensitive traveler, expense and settlement data to be accessed through controlled business components rather than directly from the user interface. Authentication and authorization can be enforced at the application boundary before operations reach the data layer. This is a design-level security advantage; the problem statement itself identifies performance and security as NFR concerns.

Performance Benefit: The settlement engine is required to execute in under 100 ms for groups of up to 50 travelers and 500 expenses. A layered design avoids unnecessary distributed-service network overhead and keeps the min-flow calculation in a dedicated business component, making it easier to optimize and benchmark against the stated latency target.

Components and Interfaces: The diagram contains User Interface, Group & Trip Management, Expense Management, Currency Conversion, Settlement Engine and Database components, with a Daily Exchange Rate Provider shown as an external dependency. Interfaces cover group/trip data, expense data, FX conversion, daily exchange rates, balance data, settlement requests and database operations.

Conclusion: Layered Architecture is appropriate because it provides clear separation of concerns while directly supporting the problem's core workflow: record uneven expenses, normalize currencies using daily rates, calculate net balances, and generate the minimum-transaction settlement graph.

UML COMPONENT DIAGRAM

Layered Architecture

